

Qatar-1b

R-1162

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_241103 · created 2026-07-31T02:31:24+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460617.7161 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.039
Transit depth (Rp/Rs)^2	2.65 +/- 1.27 [%]
Orbital Inclination (inc)	83.04 +/- 2.1
Airmass coefficient 1 (a1)	1.083 +/- 0.016
Airmass coefficient 2 (a2)	-0.039 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	#3 - [333, 95]
Optimal Method	PSF photometry
Transit Duration (day)	0.053 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.163 ± 0.039 vs 0.1463 (deviation 0.42σ)
Duration fit vs published	0.053 d vs 0.0692 d (deviation 0.98σ)
Mid-time O–C	-3.6 min
Depth SNR	4.1
Residual scatter	1.49 %

Light curve

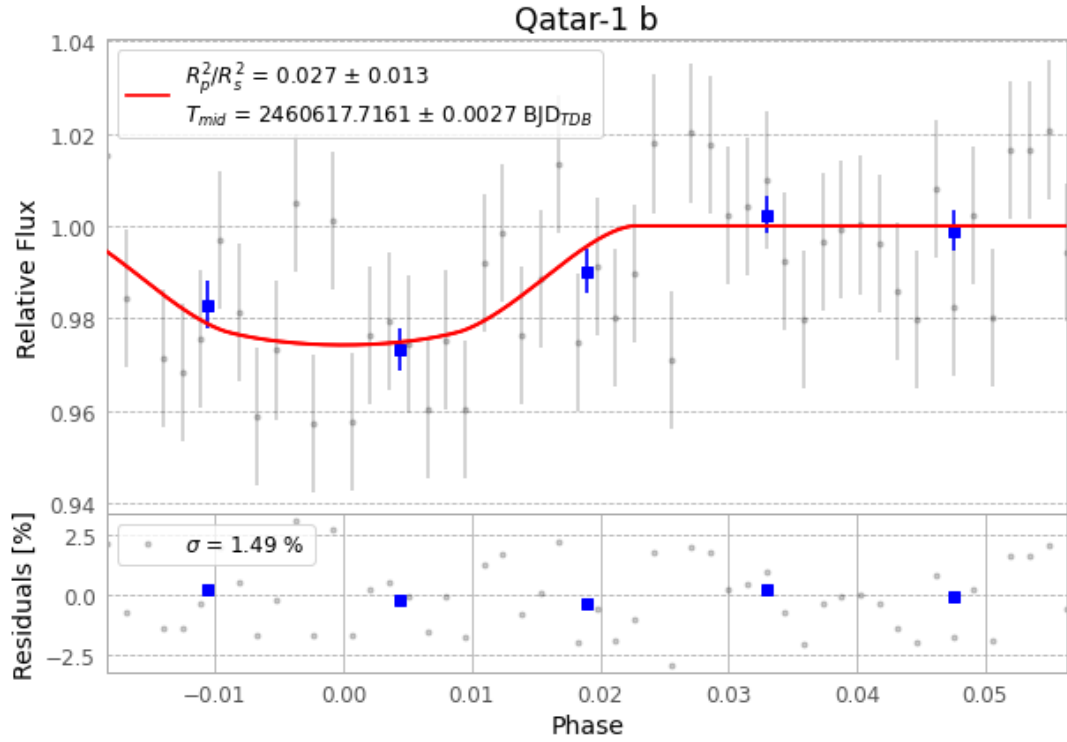


Figure 1 — FinalLightCurve_Qatar-1 b_2024-11-02.png (SHA-256 160d4201b65f573c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 289], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf8a29e698bc1fb8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

52 FITS frames (34 MB), Qatar-1241103043016.FITS → Qatar-1241103070315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-241103.log (66541dac598a...), FinalLightCurve_Qatar-1 b_2024-11-02.png (160d4201b65f...), FinalLightCurve_Qatar-1 b_2024-11-02.csv (52fe395388f9...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:31Z.